

Presentation speech

Speaker: Afnan · about 5½ minutes of speaking, 6–7 minutes with pauses and clicks · one bullet = one sentence to say · bold = the key word or number to remember · ► brackets = what to do, not what to say

Every number here is taken from the thesis report as submitted. Where the slide wording and the report disagree, the speech follows the report.

SLIDE 1 Title

0:42

- Good morning, respected members of the panel.
- We are group **P25301051**.
- I am Afnan Mazumdar, with Jerin Aktar and Shoyeb Hasan Sayem.
- Our supervisor is **Utsho** Kumar Roy sir.
- We built a **three-model pipeline** on the CUAD dataset.
- It **finds** the important clauses in a contract, **extracts** their text, gives each a **risk level**, and adds a **plain-English** sentence.
- Everything is tested on contracts the models **never saw**.
- **Three of our five research questions came back "no."**
- We present those negative results as our **contribution**, because each one corrects what the component scores alone would suggest.

► click

SLIDE 2 The problem, and the data

1:02

- Contracts are **long**, and the clauses that matter are **short**.
- The median contract is about **33,000 characters**.
- The median clause is only **196 characters**.
- So a contract is about **sixteen times** longer than what a transformer can read at once.
- CUAD has **510 real contracts**, with lawyers marking clauses in **41 categories**.

► point to the cards

- We split it **80/20 at the contract level**, with seed 42.
- That gives **408 contracts to train** and **102 to test**.
- All three models use the **same split**, so no training clause can reach the test set.
- The data is **severely imbalanced**, from 100 percent down to 2.5 percent.

- **Five categories** with seven or fewer test positives score **zero**.
- And the data **cannot be grown**: annotation cost about **two million dollars**.
- So **transfer learning** is mandatory, and every result is on **held-out** contracts.

► click

SLIDE 3 The pipeline, and how it reads a contract

1:08

- Our pipeline has **four stages**.
- **Presence**: TF-IDF and a windowed DistilBERT decide which clauses are there.
- **Span**: DistilBERT-QA extracts the clause text.
- **Risk**: a per-category lookup we wrote from the **weaker party's** side.
- **Explanation**: FLAN-T5 adds a one-line sentence.

► point to the red and green boxes

- Measured end to end, presence loses **22 percent** of clauses.
- Span loses another **56.3 percent**.
- Risk loses **nothing**, because it is a lookup.
- The explanation picks the **wrong template 30 percent** of the time.

► point to the window diagram

- We slide a **2,000-character window** in **1,500-character steps**.
- The **500-character overlap** is longer than the median clause.
- We score every window and take the **maximum**.
- If **any window fires**, the clause is present.
- This **removes the recall ceiling**.
- But it **costs precision**, because one wrong window flips the whole contract.
- Retrieval could find the right window only **64 percent** of the time, below the baseline.
- So we **dropped retrieval** before training.

► click

SLIDE 4 How we trained, and how we test

0:50

- Everything trained on one **Apple M4 laptop** with **16 gigabytes**, no CUDA.

► point to the rows

- Presence took about **49 minutes**, span **23**, and explanation **8**.
- DeBERTa gave **NaN gradients** on the Mac, so we used **DistilBERT**.

► point to the cards

- **No hyperparameter was ever tuned.**
- Only presence had a small validation set: **5 percent of training windows**, for monitoring.
- That **protects the test set**, but the transformer competed **untuned**.
- We report **one mistake**: an early summarizer run monitored loss on **300 test clauses**.
- It selected **no checkpoint**, and the reported model was **rebuilt without it**.
- We test on **102 unseen contracts**, **4,182 decisions** and **2,667 span windows**.
- Our bootstrap resamples **whole contracts, not cells**.

► click

SLIDE 5 RQ1: the transformer loses to a bag of words

0:48

- Our first question: can a windowed transformer beat a full-document baseline?
- The answer is **no**.

► point to the bar chart

- The **TF-IDF baseline** scores **0.775**.
- The **transformer** scores **0.694**.
- The baseline wins by **0.081**, in **every one** of ten thousand bootstrap resamples.
- TF-IDF reads the **whole contract**, and max-pooling creates **false positives**.
- The **AND ensemble** scores **0.776**, but we **withdrew** that win.
- Its lead is only **0.0014**, and its confidence interval **crosses zero**.
- So it is a **tie**.

► point to the small chart

- Formulaic categories are nearly solved: **Document Name 1.000**, **Parties 0.995**, **Governing Law 0.961**.
- Rare categories score **zero**.
- That is why the **baseline's macro-F1** is only **0.572**.

SLIDE 6 We ranked a leaderboard by noise

0:52

- We tested **every excuse** for the transformer.

► point to the bullets

- **Undertrained?** Training on all 53,662 windows changed the score by **minus 0.0004**.
- **Unlucky seed?** Across three seeds, the gap is **9.5 standard deviations**.
- **Wrong pooling?** We tried six rules, and **none reaches the baseline**.
- **Bad threshold?** It explains **0.056**, a real caveat, but still not enough.
- So **the transformer loses**.
- It is **not** a budget, seed or aggregation artefact.

► point to the chart

- The ensemble's lead is only **0.0014**.
- Seed noise is **0.0022**.
- Split variance is **0.0063**.
- The bootstrap half-width is **0.0078**.
- **Every error bar is bigger than the result**.
- We keep the ensemble for its **balanced precision and recall**.
- It is also **four times less sensitive** to the random seed.

► click

SLIDE 7 The threshold, and what the confidence means

0:57

- Two things we measured **after the fact**.

► point to the left plot

- **First, the threshold**.
- **0.5** is just the softmax default.
- At **0.9**, the transformer reaches **0.750** instead of **0.694**.
- But we **do not adopt it**.
- Choosing a threshold on **test data** is exactly the tuning our protocol **forbids**.

► point to the right plot

- **Second, the confidence.**
- The calibration error is **0.201**.
- A score above 0.9 means the clause is there only about **73 percent** of the time.
- So our app's confidence bar **overstates certainty**.
- The model also **over-fires**: precision is only **0.562**.
- A maximum over about **35 windows** is **35 chances to be wrong**.

► point to the green card

- The fix: **carve a validation split** from the training contracts.
- Choose the threshold there, and **calibrate** the score there.
- Now **[teammate's name]** will show what these errors mean for the high-risk clauses.

► hand over

Three things the speech says differently from the slides

- **Slide 1:** say "Utsho Kumar Roy" (the slide spells it "Utsha").
- **Slide 5:** say "the **baseline's** macro-F1 is 0.572". The slide puts 0.572 next to the ensemble's 0.776, but it belongs to the TF-IDF baseline (report §5.3).
- **Slide 7:** say "about 35 windows". The slide prints "-35".

Backup answer: "Does the presence model read the whole 2,000-character window?"

Only use this if someone asks. The code truncates each (question, window) input at 256 tokens with no stride, so the model reads roughly the **first 1,000 characters** of each window.

- **No.** At 256 tokens, it reads about the **first half** of each window.
- That is **not discussed** in the report.
- It is likely **one more reason** the transformer loses to TF-IDF, which reads the whole contract.
- None of our four controls changes the windowing, so they **can't rule it out**.
- The fix is cheap: **~900-character windows**, with an overlap still longer than the median clause.